# Product Requirements Document (PRD)

## Sea Lion Education

### Offline-first educational AI platform for Indonesian schools in outermost and underdeveloped regions

**Version:** 1.0 (MVP)

**Status:** Draft

**Target release:** End of 2026

**Audience:** Engineers, UI/UX designers, researchers, university stakeholders, and implementation partners

# Executive summary

Sea Lion Education is a smart-client educational AI platform designed for Indonesian schools in outermost, underdeveloped, and resource-constrained regions where internet access is unavailable or unreliable. The platform runs on a local school device and can optionally connect to cloud services when internet access is available. **Elise**, short for E-Library Sea Lion, serves K–12, junior high, and senior high school students by enabling interactive learning through an instant messaging-style interface, with an initial focus on Indonesian and English language learning.

The MVP delivers a lightweight conversational learning experience powered by a Sea Lion Small Language Model (SLM) and a Retrieval-Augmented Generation (RAG) system built from Indonesian curriculum books. It supports three core learning modes: Ask, Learn, and Practice.

1. Ask mode: its simples conversation mode such as “Beri contoh soal hukum Newton 2?”. The AI will work to answer the question.
2. **Learn mode:** Students engage in a guided conversation to understand a selected topic, such as “Saya ingin belajar bab gradien di matematika kelas 2.” The AI supports learning by asking follow-up questions, explaining concepts, and helping students achieve the intended learning outcomes.
3. **Practice mode:** Students practice for assessments such as daily quizzes, midterms, finals, or national exams in a specific subject. For example, a student may request, “Saya ingin menyiapkan diri untuk berlatih ulangan harian termodinamika.” The AI generates multiple-choice, short-answer, or essay questions, evaluates the responses, calculates a score, and provides improvement recommendations.

The primary objective of the MVP is to validate that a locally deployed AI education platform can run smoothly on limited hardware while delivering reliable, curriculum-grounded learning support for students. **Elise** is smart client platform which means it can work online and offline.

# Problem statement

Students in many Indonesian outermost, underdeveloped, and resource-constrained regions have limited access to reliable learning support, qualified teaching assistance, and AI-powered educational tools. Although curriculum books may be available, students often lack an interactive system that can help them ask questions, understand topics, and practice assessments when internet connectivity is unavailable or unstable.

The core problem is the absence of accessible, curriculum-grounded learning assistance that can operate locally on limited school hardware while still allowing optional cloud connectivity for content updates and service enhancement when internet access is available.

Elise addresses this gap by bringing Sea Lion Education directly into schools as a smart-client educational AI platform. The system runs on a local school device, stores curriculum knowledge locally, supports offline operation, and can optionally connect to cloud services to update learning materials or improve the knowledge base when connectivity is available.

# Vision

To make curriculum-grounded AI learning accessible to every Indonesian student by delivering Elise as a smart-client educational platform that works offline on local school devices, supports Ask, Learn, and Practice interactions, and can connect to cloud services when available without making internet access a dependency.

# MVP goal

## Primary goal

Build Elise as a smart-client educational AI platform that enables students to ask questions, learn topics, and practice assessments using Indonesian curriculum books through a simple conversational interface. The MVP must run on a local school device, work without internet dependency, and optionally connect to cloud services when connectivity is available.

## Success criteria

The MVP is considered successful if:

1. The platform runs reliably on a **local school device or cloud-based server**.
   - Students can complete **Ask**, **Learn**, and **Practice** workflows.
   - The AI remains grounded in the curriculum knowledge base.

2. Optional cloud connectivity can **support content updates** and **buy service enhancement** without becoming required dependency.

# Goals and non-goals

## Goals

1. Smart-client operation with offline-first capability
2. Local school device and local network accessibility
3. Curriculum-grounded AI responses
4. Grade-based learning personalization
5. Practice generation and grading
6. Teacher monitoring dashboard
7. Knowledge base management
8. Optional cloud connectivity for curriculum updates and knowledge base enhancement

## Non-goals

1. Personalized learning roadmaps
2. Internet-dependent learning features Mobile application
3. Mandatory online synchronization during learning sessions
4. Collaborative classrooms
5. Assignment management
6. Payment or subscription systems
7. Advanced analytics
8. Multi-school deployment through a shared central server
9. Teacher role

# Target users

## Primary user: Student

**Profile**

- Kindergarten to senior high school
- Limited technical literacy
- Using the system through a school computer or BYOD (Bring your own device)
- Requires simple and intuitive interactions

**Primary goals**

- Understand school subjects
- Interactive learning guide
- Practice and improve learning outcomes

# Product overview

Elise, short for E-Library Sea Lion, is the student-facing product within Sea Lion Education. It provides a smart-client conversational learning experience that runs on a local school device, supports offline use, and can optionally connect to cloud services when internet access is available. The platform includes three core learning modes: Ask, Learn, and Practice.

## Ask mode

Ask mode allows students to ask direct questions about a subject or topic using a simple instant messaging-style interface.

The AI retrieves relevant content from the curriculum knowledge base and responds with grounded explanations, examples, or clarifications. If the answer cannot be supported by available materials, the AI clearly indicates that the information is not available from the current knowledge base.

## Learn mode (Guided-Learning Mode)

Learn mode provides guided learning conversations for students who want to understand a selected topic step by step.

Students can state what they want to learn, such as a subject, or topic. The AI then explains the topic, asks follow-up questions, checks understanding, and helps the student progress toward the intended learning outcomes.

## Practice mode

Practice mode helps students prepare for assessments such as daily quizzes, midterms, finals, or national exams by generating exercises from the curriculum knowledge base just like NotebookLM user experience when generating Quiz.

- Topic
- Question type
- Number of questions

Supported question types:

- Multiple choice
- Short answer
- Essay

The AI supports multiple-choice, short-answer, and essay questions. It evaluates student responses, explains correctness, calculates a final score, and provides recommendations for improvement based on topic mastery.

# User flow

## Student onboarding

1. Register account
2. Create username and password
3. Select current grade
4. Enter platform

The selected grade determines the default curriculum knowledge available in Learn mode.

## Ask flow (Normal Conversation)

1. Open Learn
2. Ask educational question
3. AI retrieves **grade-specific** curriculum content
4. AI provides conversational explanation
5. Conversation is saved

## Learn flow (Guided-Learning)

1. Open Learn
2. Select guided-learning mode
3. Select course
4. Define topic
5. AI retrieves **grade-specific** curriculum content
6. AI will ask educational question to train our ability on a certain topic
7. AI provides conversational explanation
8. Conversation is saved

## Practice flow

1. Open Practice
2. Select course
3. Define specific topic
4. Select question type
5. Select number of questions
6. Answer questions
7. Receive grading
8. View score and mastery summary
9. Session is saved

# Functional requirements

## Authentication

### Student

- Register
- Login
- Logout
- Change password

## Grade management

Students must have a grade associated with their account.

The system uses this grade to determine available curriculum materials.

Teachers can update student grades if necessary.

## Ask mode (Normal Conversation)

The system shall:

- Search all curriculum books for the student’s grade
- Maintain conversation context within the session
- Provide educational explanations
- Adapt language complexity to the student’s grade
- Store conversation history

## Learn mode (Guided-Learning)Learn mode (Guided-Learning)

The system shall:

- Search all curriculum books for the student’s grade
- Maintain conversation context within the session
- Train student ability via active conversation (ask -> explain -> loop)
- Adapt language complexity to the student’s grade
- Store conversation history

# Practice mode

The system shall:

- Generate questions from some topic
- Support MCQ, short answer, and essay
- Automatically evaluate responses
- Explain correct and incorrect answers
- Calculate final scores
- Produce topic mastery summaries
- Save practice history

# Conversation history

Conversation history shall be separated by mode.

Students can:

- View history
- Rename conversations

Conversations persist across logins.

# Progress tracking

Students shall have a progress page displaying:

- Practice history
- Scores
- Topic mastery
- Recent activity

# AI behavior requirements

## Retrieval grounding

All educational responses must be grounded in the curriculum knowledge base.

## Hallucination handling

If sufficient information is unavailable, the AI must explicitly state that the answer cannot be determined from the available books.

## Language adaptation

The AI shall adjust:

- Vocabulary
- Explanation complexity
- Sentence structure

based on the student's grade level.

## Tutor personality

The AI should behave as a friendly and encouraging tutor rather than a formal lecturer.

## Safety

The AI must refuse:

- Harmful requests
- Inappropriate content
- Dangerous instructions
- Non-educational unsafe interactions

# AI Capability Requirements

## Guided Learning

- The LLM must be able to operate in guided learning mode -> proactively asking questions to build the user's understanding of a specific topic.

## Grounded Answer

- The LLM must be able to answer only based on the context given and refused to answer if there are no context given.

## MCQ Generation

- The LLM must be able to generate multiple-choice questions (MCQs) in a **structured output format**.
- Input: Topic
- Output: MCQ + Answer

## Essay/Short Answer Generation

- The LLM must be able to generate an essay or short answer in a **structured output format**.
- Input: Topic
- Output: Question

## Essay/Short Answer Evaluation

- The LLM must be able to evaluate an essay or short answer in a **structured output format**.
- Input: Question and Student Answer
- Output: Correct/Wrong Answer and Explanation

# Error handling

## AI failure

- Retry request
- Display friendly error message

## Unsupported knowledge

Display:

“I don’t know based on the available books.”

## Excessive input

Limit message length before submission.

# Product roadmap

## Version 1.0 – Student platform (MVP)

Focus: curriculum-grounded AI learning for students.

Features:

- student authentication,
- grade onboarding,
- Ask mode,
- Guided Learning mode,
- Practice mode,
- conversation history,
- progress tracking,
- Offline Mode,
- and Online Mode.

## Version 2.0 – Extended student platform

Focus: enhanced learning experience.

Planned features:

- personalized learning recommendations,
- adaptive practice difficulty,
- extended learning content,
- additional educational resources,
- and improved AI tutoring capabilities.

## Version 3.0 – Teacher and school platform

Focus: school administration and learning monitoring.

Planned features:

- teacher authentication,
- teacher dashboard,
- student progress monitoring,
- conversation review,
- practice analytics,
- weak topic identification,
- knowledge base management,
- curriculum content upload,
- and school-level administration.

# MVP acceptance criteria

The MVP is complete when:

1. Students can register and log in.
2. Students can select and update their grade.
3. Ask mode retrieves grade-specific curriculum knowledge.
4. Guided Learning provides structured instructional conversations.
5. Practice mode generates and grades exercises.
6. Scores and topic mastery summaries are stored.
7. Conversation history persists across sessions.
8. Progress tracking displays student learning activity.
9. The system operates fully in Offline Mode.
10. The system operates fully in Online Mode.
11. The learning experience is functionally consistent across both deployment modes.